

Ward score – winrate

Kaggle- <https://www.kaggle.com/datasnaek/league-of-legends> (data from 51490 games)

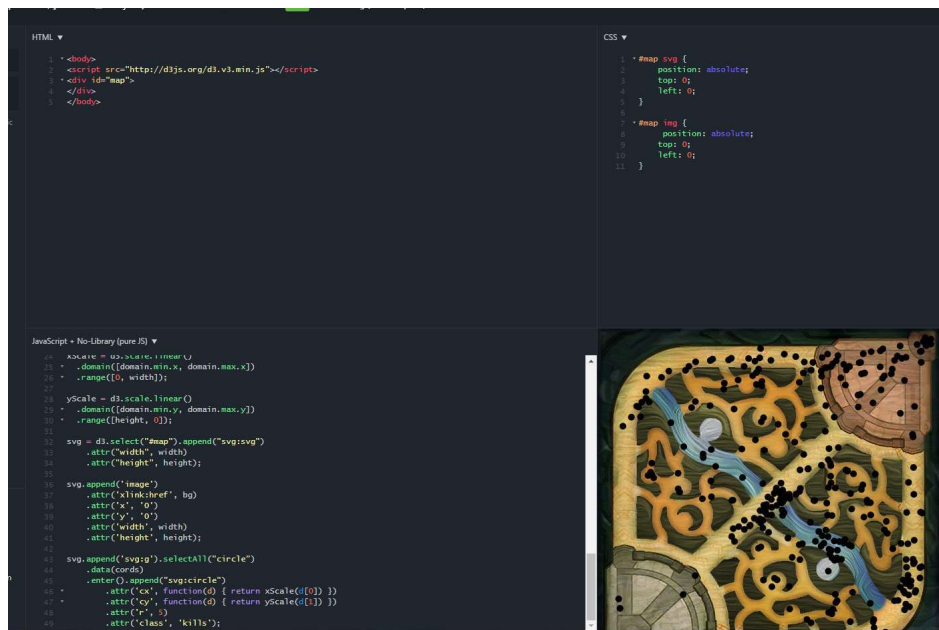
<https://www.kaggle.com/jaytegge/league-of-legends-data-analysis>

League of legend data.

Ult – winrate

Red side blue side – winrate

- Total time of CC dealt
- Number of times each ability is cast
  - All 4 abilities and summoner spells
- Level at end of game



<http://web.cs.wpi.edu/~claypool/mqp/lol-crawler/final-report.pdf>

Analysis of data from API.

Majorly analysis of winrate vs. pickrate, banrate.

Winrate about different roles. (fighter are pick the most, ban the most, most influential)

<https://medium.com/@farzatv/deepleague-leveraging-computer-vision-and-deep-learning-on-the-league-of-legends-mini-map-giving-d275fd17c4e0>

Analysis the movement of player.

<http://cs229.stanford.edu/proj2016/report/Lin-LeagueOfLegendsMatchOutcomePrediction-report.pdf>

prediction of a currently ongoing game with all of the features of that game in mind

<https://hackernoon.com/league-of-legends-predicting-wins-in-champion-select-with-machine-learning-6496523a7ea7>

Predicting win in champion select. 60% accuracy, sometime 90 percentage of a loss. Adc least impactful.

<https://towardsdatascience.com/ai-in-video-games-improving-decision-making-in-league-of-legends-using-real-match-statistics-and-29ebc149b0d0>

Try to improve player's decision making.

<http://www.trailofpapers.net/2015/10/playing-in-random-forests-in-league-of.html>

Random forest proves that 6 percent surrender is winnable.

<https://data-speaks.luca-d3.com/2017/05/machine-learning-to-analyze-league-of.html>

2017 spring split. Predicting Victory

<https://dotesports.com/league-of-legends/news/study-skill-games-lol-correlates-higher-intelligence-18755>

skills in games correlate to higher intelligence

<https://www-igi-global-com.mutex.gmu.edu/gateway/article/full-text-html/182451>

Mental issues for optimal performance.

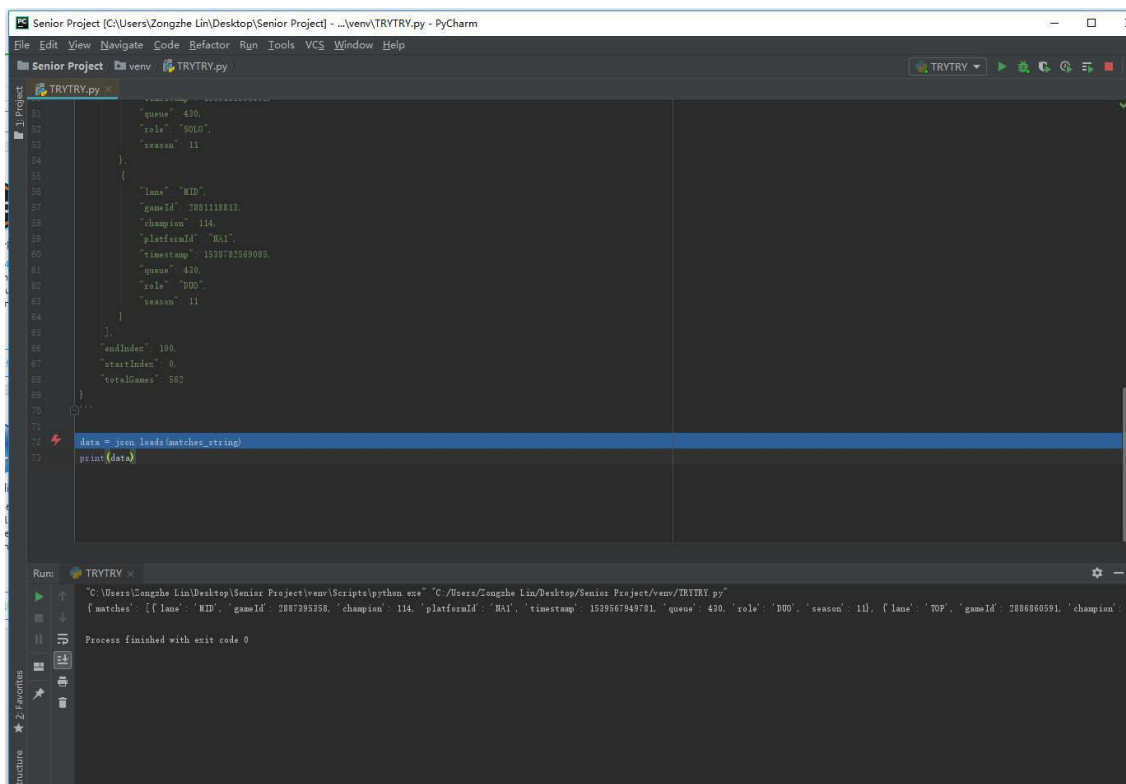
<https://compete.kotaku.com/big-data-is-the-future-of-esports-1827206481>

big data Jungle Proximity

Watson's team analyzes the proximity of the jungler to each lane throughout the course of the game, in order to quantify how good a jungler is at jungling. During games, his team tracks each champion's X and Y coordinates on the map each second. Using their exact positions over a stretch of time, they can determine their proximity to each other and begin to analyze that connection.

```
"accountId": 200769405,  
"id": 37743667,  
"revisionDate":
```

Fiora's champion id = 114



The screenshot shows a PyCharm IDE window titled "Senior Project [C:\Users\Zonghe Lin\Desktop\Senior Project] - ...\venv\TRYTRY.py - PyCharm". The main editor displays a Python script with the following content:

```
11 match_string = '''  
12     "queue": 430,  
13     "role": "SOLO",  
14     "season": 11  
15 },  
16 {  
17     "lane": "MID",  
18     "gameId": 2001110813,  
19     "champion": 114,  
20     "platformId": "NA1",  
21     "timestamp": 1538782549005,  
22     "queue": 430,  
23     "role": "DUO",  
24     "season": 11  
25 }  
26 ]  
27 "endIndex": 100,  
28 "startIndex": 0,  
29 "totalGames": 562  
30 }  
31 '''  
32  
33 data = json.loads(match_string)  
34 print(data)
```

The script is executed, and the output is displayed in the Run console:

```
Run: TRYTRY  
C:\Users\Zonghe Lin\Desktop\Senior Project\venv\Scripts\python.exe "C:\Users\Zonghe Lin\Desktop\Senior Project\venv\TRYTRY.py"  
{  
  "matches": [  
    {  
      "lane": "MID",  
      "gameId": 2007295258,  
      "champion": 114,  
      "platformId": "NA1",  
      "timestamp": 1539567949781,  
      "queue": 430,  
      "role": "DUO",  
      "season": 11,  
      "lane": "TOP",  
      "gameId": 2006040591,  
      "champion":  
    }  
  ]  
}
```

The output shows a JSON object with a "matches" array containing two match entries. The first entry is for a "MID" lane with "gameId": 2007295258, "champion": 114, "platformId": "NA1", "timestamp": 1539567949781, "queue": 430, "role": "DUO", "season": 11. The second entry is for a "TOP" lane with "gameId": 2006040591, "champion": 114, "platformId": "NA1", "timestamp": 1538782549005, "queue": 430, "role": "DUO", "season": 11.

```
File Edit View Navigate Code Refactor Run Tools VCS Window Help
Senior Project venv firstgame.py
TRYTRY.py x firstgame.py x
10593 x : 2596
10594 },
10595 "type": "CHAMPION_KILL"
10596 },
10597 {
10598 "timestamp": 1859764,
10599 "skillSlot": 3,
10600 "levelUpType": "NORMAL",
10601 "type": "SKILL_LEVEL_UP",
10602 "participantId": 1
10603 }
10604 ]
10605 }
10606 ],
10607 "frameInterval": 60000
10608 }
10609 """
10610
10611 data = json.loads(frames_string)
10612 print(data)
10613
10614 for laner in data['frames']:
10615     print(laner['timestamp'])
```

Run: firstgame x

1200338  
1260355  
1320366  
1380391  
1440412  
1500433  
1560456  
1620475  
1680482  
1740509  
1800538  
1860558

```
data = json.loads(frames_string)
print(data)

for laner in data['frames']:
    print(laner['timestamp'], laner['participantFrames'])
```

firstgame x

"C:/Users/Zonghe Lin/Desktop/Senior Project/venv/Scripts/python.exe" "C:/Users/Zonghe Lin/Desktop/Senior Project/venv/firstgame.py"

```
{
  "frames": [
    {
      "timestamp": 0,
      "participantFrames": {
        "1": {
          "totalGold": 500,
          "teamScore": 0,
          "participantId": 1,
          "level": 1,
          "currentGold": 500,
          "minionsKilled": 0,
          "dominionScore": 0,
          "position": {
            "y": 581,
            "x": 561
          }
        }
      }
    },
    {
      "timestamp": 60006,
      "participantFrames": {
        "1": {
          "totalGold": 500,
          "teamScore": 0,
          "participantId": 1,
          "level": 1,
          "currentGold": 0,
          "minionsKilled": 0,
          "dominionScore": 0,
          "position": {
            "y": 3218,
            "x": 120026
          }
        }
      }
    },
    {
      "timestamp": 120026,
      "participantFrames": {
        "1": {
          "totalGold": 520,
          "teamScore": 0,
          "participantId": 1,
          "level": 1,
          "currentGold": 20,
          "minionsKilled": 0,
          "dominionScore": 0,
          "position": {
            "y": 599,
            "x": 180039
          }
        }
      }
    },
    {
      "timestamp": 180039,
      "participantFrames": {
        "1": {
          "totalGold": 643,
          "teamScore": 0,
          "participantId": 1,
          "level": 1,
          "currentGold": 143,
          "minionsKilled": 0,
          "dominionScore": 0,
          "position": {
            "y": 3807,
            "x": 240059
          }
        }
      }
    },
    {
      "timestamp": 240059,
      "participantFrames": {
        "1": {
          "totalGold": 1036,
          "teamScore": 0,
          "participantId": 1,
          "level": 2,
          "currentGold": 536,
          "minionsKilled": 0,
          "dominionScore": 0,
          "position": {
            "y": 10076,
            "x": 300073
          }
        }
      }
    },
    {
      "timestamp": 300073,
      "participantFrames": {
        "1": {
          "totalGold": 1244,
          "teamScore": 0,
          "participantId": 1,
          "level": 3,
          "currentGold": 744,
          "minionsKilled": 0,
          "dominionScore": 0,
          "position": {
            "y": 6574,
            "x": 360094
          }
        }
      }
    },
    {
      "timestamp": 360094,
      "participantFrames": {
        "1": {
          "totalGold": 1801,
          "teamScore": 0,
          "participantId": 1,
          "level": 4,
          "currentGold": 1301,
          "minionsKilled": 1,
          "dominionScore": 0,
          "position": {
            "y": 11103,
            "x": 420117
          }
        }
      }
    },
    {
      "timestamp": 420117,
      "participantFrames": {
        "1": {
          "totalGold": 1943,
          "teamScore": 0,
          "participantId": 1,
          "level": 4,
          "currentGold": 193,
          "minionsKilled": 1,
          "dominionScore": 0,
          "position": {
            "y": 2850,
            "x": 120026
          }
        }
      }
    }
  ]
}
```

2887395358  
 2886860591  
 2885756007  
 2885241983  
 2883958910  
 2881118813

Game 1: We win no.8 True, True  
 Game 2: we win-high xp no.9 True, True  
 Game 3: we win-high xp no.3 True, True  
 Game 4: we win - no.4 True, False - drg ,rift, 2 barons  
 Game 5: no.8 True, True  
 Game 6: no.4 True, True

#### Lopez

2903840921 - 7: no.5 False, True  
 2903950066 - 8: no.5 False, False (1 baron, rift )  
 2903183612 - 9 : no.5 False, True  
 2903181534 - 10: no.3 False, False (1 baron,rift)

#### Scarlix

2903007085 - 11: no.2 False,False (first drg, rift)  
 2901982542 - 12: no.9 False, True  
 2903007085 - 13: no.2 False, False (first drg, rift)  
 2901982542 - 14: no.9 False, True  
 2901917945 - 15: no.5 False, True  
 2901796335 - 16: no.9 True, True  
 2901766151 - 17: no.9 True, True  
 2901684844 - 18: no.1 False, False (rift)  
 2901658661 - 19: no.3 True, True  
 2901640990 - 20: no.2 False, True  
 2901453565 - 21: no.2 False, False (first drg, rift)  
 2901402685 - 22: no.3 False, False (first drg,rift,baron)  
 2901345200 - 23: no.2 False, True

#### Best Tryndora NA

2894190026 - 24: no.1 False, True  
 2894183710 - 25: no.3 False, False (nothing)  
 2894035756 - 26: no.7 False, False (three towers, FF)  
 2893520309 - 27: no.1 False, False (baron, drg, rift)  
 2893425862 - 28: no.10 False, False (first rift, --first baron(2), baron,first inhib)  
 2893302644 - 29: no.3 False, False(first tower, rift)  
 2893165654 - 30: no.4 False, False(first rift,first inhib)

First check(if fiara has highest exp)

Second check(if win)

14/30 (14 losses-16 wins)

8/8(highest xp always win)

(12/14 losses) lose rift

(5/14 losses) baron

